

Question 1 (12 marks)

Marks

a) Differentiate the following;

i. e^{-3x} (1)

ii. $e^{\sin x}$ (1)

iii. $\log_e (4x+1)^2$ (2)

iv. $\frac{\log_e x}{x}$ (2)

b) Find the following;

i) $\int e^{-2x} dx$ (1)

ii) $\int_1^6 \frac{6}{2x+3} dx$ (2)

c) Find the equation of the normal to the curve $y = e^{x^2}$ at $(1, e)$. (3)**Question 2** (12 marks) (START A NEW PAGE)a) Draw a sketch of $y = 3 \cos 2x$ over $0 \leq x \leq 2\pi$ (2)

b) Differentiate the following;

i. $y = \tan^2 4x$ (2)

ii. $y = 2 \sin x^2$ (2)

iii. $y = e^{2x} \cos 3x$ (2)

c) Find the area between the two curves $y = \cos x$ and $y = \sin x$ over the domain (2)

$$0 \leq x \leq \frac{\pi}{4}.$$

d) Find the volume of the solid formed by $y = \sec x$ being rotated about the x axis (2)

$$\text{between } x = 0 \text{ and } x = \frac{\pi}{4}.$$

Question 3 ... page 2

Question 3 (11 marks) (START A NEW PAGE)

- a) A piece of plyboard has many slits cut into it. Each pair of slits has a constant difference in length. The shortest slit is 30cm and the longest slit is 50cm.
- If the total length of all the slits is 1240cm, find the number of slits. (2)
 - Find the constant difference in length between the slits. (2)
 - Find the length of the 11th shortest slit. (1)
- b) In a geometric sequence of numbers the third term is 72 and the eighth term is 17496.
- Find the first term and the common ratio. (2)
 - Which is the largest term of the sequence less than 500000? (2)
 - What is the sum of the terms less than 500000? (2)

Question 4 (10 marks) (START A NEW PAGE)

- a) Frank and Eddie shared a Lotto win. Their shares are \$10000 each.
- Frank invested his \$10000 in an investment account on January 1st 2000. It is invested at 9% per annum, compounded monthly. How much would be in the investment account on January 1st 2010, if no other payments were made during this period? (2)
 - Eddie invested his \$10000 in an investment account on the same day as Frank. The interest on his investment account was compounded annually, at 9% per annum. He decided to deposit an extra \$1000 on January 1st, each year from January 1st 2001 to 1st January 2010. How much would be in his account on January 1st 2010, including the last deposit. (4)
- b) Jan borrows \$8000 and agrees to repay it in equal instalments each year for 10 years. If the interest is charged at 7% per annum on any part of unpaid debt, calculate the amount of each yearly instalment. (4)
- Let $\$A_n$ be the amount owed on the loan after n years and let M be the instalments.
Show that $A_n = 8000(1.07)^n - M \left(\frac{1.07^n - 1}{1.07 - 1} \right)$
 - Hence, calculate the amount of each yearly instalment.

END OF TEST

Question 1 (12 marks)

(a)(i) $\frac{d}{dx} e^{-3x} = -3e^{-3x} \checkmark (1)$

(ii) $\frac{d}{dx} e^{\sin x} = \cos x e^{\sin x} \checkmark (1)$

(iii) $\frac{d}{dx} \log_e (4x+1)^2 = 2 \log_e (4x+1)$
 $\frac{d}{dx} = 2x \frac{4}{4x+1} \checkmark$

$= \frac{8}{4x+1} \checkmark (2)$

(iv) $\frac{d}{dx} \frac{\log_e x}{x} = \frac{x \times \frac{1}{x} - \log_e x \cdot 1}{x^2} \checkmark$

$= \frac{1 - \log_e x}{x^2} \checkmark (2)$

(b)(i) $\int (e^{-x}) dx = -e^{-x} + C \checkmark (1)$ (do not deduct marks for missing +C)

(ii) $\int_1^6 \frac{6}{2x+3} dx = \int_1^6 3x \frac{2}{2x+3} dx$

$= [3x \log_e (2x+3)]_1^6 \checkmark$

$= 3 (\log 15 - \log 5)$

$= 3 \log \frac{15}{5}$

$= 3 \log 3 \checkmark (2)$

$= \log 27$

(c) $y = e^{x^2}$

$\frac{dy}{dx} = 2x e^{x^2} \checkmark$

gradient of tangent at (1, e)

$\therefore m = 2e$

gradient of normal

$m = -\frac{1}{2e} \checkmark$

equation of normal

$y - e = -\frac{1}{2e} (x - 1) \checkmark (3)$

$2ey - 2e^2 = -x + 1$

$\therefore x + 2ey - 2e^2 - 1 = 0$

Question 2 (12 marks)

a) $y = 3 \cos 2x \quad 0 \leq x \leq 2\pi$

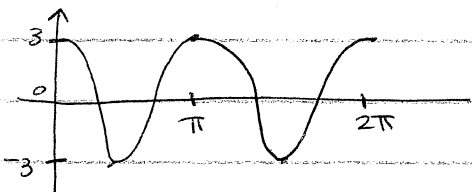
amplitude = 3

period = π

1 mark correct curve

1 correct amplitude and period.

(2)



b) (i) $y = \tan^2 4x$

$$\frac{dy}{dx} = 2 \tan 4x \cdot 4 \sec^2 4x$$

$$= 8 \tan 4x \cdot \sec^2 4x \quad (2)$$

(iii) $y = 2 \sin x^2$

$$\frac{dy}{dx} = 2 \cdot 2x \cos x^2$$

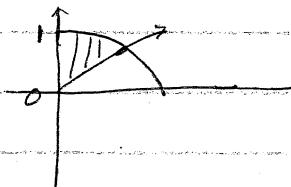
$$= 4x \cos x^2 \quad (2)$$

(ii) $y = e^{2x} \cdot \cos 3x$

$$\frac{dy}{dx} = e^{2x} \cdot -3 \sin 3x + \cos 3x \cdot 2e^{2x}$$

$$= -3e^{2x} \sin 3x + 2e^{2x} \cos 3x \quad (2)$$

c) $A = \int_0^{\pi/4} \cos x - \sin x \, dx$



$$= [\sin x + \cos x]_0^{\pi/4}$$

$$= \left(\sin \frac{\pi}{4} + \cos \frac{\pi}{4} \right) - (\sin 0 + \cos 0)$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1 \quad (2)$$

$$= \frac{2}{\sqrt{2}} - 1$$

$$= \sqrt{2} - 1$$

d) $V = \pi \int_a^b y^2 \, dx$

$$= \pi \int_0^{\pi/4} \sec^2 x \, dx$$

$$= \pi [\tan x]_0^{\pi/4}$$

$$= \pi (\tan \frac{\pi}{4} - \tan 0)$$

$$= \pi \cdot 1 \quad (2)$$

Question 3

a) $a = 30\text{cm}$ $l = 50\text{cm}$

(i) $S_n = 1240\text{cm}$

$$S_n = \frac{n}{2}(a+l)$$

$$1240 = \frac{n}{2}(30+50) \quad \checkmark$$

$$1240 = \frac{n}{2}(80)$$

$$1240 = n \times 40$$

$$n = 31$$

$$\checkmark (2)$$

$\therefore$ there are 31 slits.

(ii) $T_{31} = 50$

$$a + (n-1)d = 50$$

$$30 + 30d = 50 \quad \checkmark$$

$$30d = 20$$

$$d = \frac{2}{3}\text{cm}$$

$$\checkmark (2)$$

$\therefore$ the constant difference is $\frac{2}{3}\text{cm}$

(iii) $T_{11} = 30 + 10 \times \frac{2}{3}$

$$= 36\frac{2}{3}\text{cm}$$

$$\checkmark (1)$$

b) $T_3 = 72 = ar^2 \quad \text{--- (1)}$

$$T_8 = 17496 = ar^7 \quad \text{--- (2)}$$

$$243 = r^5$$

$$3 = r \quad \checkmark$$

$$\therefore a \times 3^2 = 72$$

$$a = 8 \quad \checkmark$$

$$(2)$$

$\therefore a = 8$ and $r = 3$

c) $ar^{n-1} < 500000$

$$8 \times 3^{n-1} < 500000 \quad \checkmark$$

$$3^{n-1} < 62500$$

$$n-1 \log 3 < \log 62500$$

$$n-1 < \frac{\log 62500}{\log 3}$$

$$n-1 < 10.05170063$$

$$n-1 = 10$$

$$n = 11$$

$$(2)$$

$\therefore$ the 11th term is

the largest less than 500 000

Question 3

$$\begin{aligned} b)(iii) \quad S_n &= \frac{a(r^n - 1)}{r - 1} \\ &= \frac{8(2^{11} - 1)}{(2 - 1)} \checkmark \\ &= 708587 \checkmark (2) \end{aligned}$$

∴ The sum of terms 708587

Question 4

(a) (i) $P = \$10000$

$$r\% = 9\% \text{ p.a.} = 0.75\% \text{ per month} = 0.0075$$

$$n = 10 \text{ years} = 120 \text{ months}$$

$$A = 10000 (1.0075)^{120} \quad \checkmark$$

$$= \$24513.57078 \quad \checkmark \quad (2)$$

$$= \$24513.57 \text{ (n.c.)}$$

(ii) $P = \$10000 + \$1000 \text{ each year after.}$

$$r\% = 9\% \text{ p.a.} = 0.09$$

$$n = 10 \text{ years}$$

$$2000 \quad A_1 = 10000 (1.09)^0$$

$$2001 \quad A_2 = 10000 (1.09)^1$$

$$2002 \quad A_3 = 10000 (1.09)^2$$

$$\vdots$$
$$2009 \quad A_{10} = 10000 (1.09)^9$$

$$2010 \quad A_{11} = 10000 (1.09)^{10}$$

$$A_{\text{TOTAL}} = 10000 (1.09)^0 + 10000 (1.09)^1 + 10000 (1.09)^2 + \dots + 10000 (1.09)^9 + 10000 \quad \checkmark$$

$$= 10000 (1.09)^0 + 10000 (1 + 1.09 + 1.09^2 + \dots + 1.09^9)$$

$$= 10000 (1.09)^0 + 10000 \left(\frac{1(1.09)^{10} - 1}{1.09 - 1} \right) \quad \checkmark$$

$$= \$38866.56646$$

$$= \$38866.57 \text{ (n.c.)} \quad \checkmark \quad (3)$$

Alternative Method

$$2000 \quad A_1 = 10000 (1.09)^0$$

$$2001 \quad A_2 = (10000 (1.09) + 10000) (1.09)^1$$
$$= 10000 (1.09)^2 + 10000 (1.09)^1$$

$$2002 \quad A_3 = (10000 (1.09)^2 + 10000 (1.09)^1 + 10000) (1.09)$$
$$= 10000 (1.09)^3 + 10000 (1.09)^2 + 10000 (1.09)$$

$$\vdots$$
$$2009 \quad A_{10} = 10000 (1.09)^0 + 10000 (1.09)^1 + \dots + 10000 (1.09)^9$$

$$2010 \quad A_T = 10000 (1.09)^0 + 10000 (1.09)^1 + \dots + 10000 (1.09)^9 + 10000 \quad \checkmark$$

$$A_T = 10000 (1.09)^0 + 10000 \left(\frac{1(1.09)^{10} - 1}{1.09 - 1} \right) \quad \checkmark$$

$$A_T = \$38866.57 \text{ (n.c.)} \quad \checkmark \quad (3)$$

$$(b) P = \$8000$$

$$r = 7\% \text{ p.a.} = 0.07$$

$$n = 10$$

$m = \text{yearly instalment.}$

$$A_1 = 8000(1.07)^1 - m$$

$$\begin{aligned} A_2 &= (8000(1.07)^1 - m)(1.07) - m \\ &= 8000(1.07)^2 - m(1.07) - m \end{aligned}$$

$$\begin{aligned} A_3 &= (8000(1.07)^2 - m(1.07) - m)(1.07) - m \\ &= 8000(1.07)^3 - m(1.07)^2 - m(1.07) - m \end{aligned}$$

$\vdots$

$$\begin{aligned} A_{10} &= 8000(1.07)^{10} - m(1.07)^9 - m(1.07)^8 - m(1.07)^7 - \dots - m \checkmark \\ &= 8000(1.07)^{10} - m(1 + 1.07 + 1.07^2 + \dots + 1.07^9) \\ &= 8000(1.07)^{10} - m \frac{(1.07^{10} - 1)}{1.07 - 1} \checkmark \end{aligned}$$

$$= 8000(1.07)^{10} - m(13.81644796) \checkmark$$

$$m = \frac{8000(1.07)^{10}}{13.81644796}$$

(4)

$$m = \$1139.020022$$

$$m = \$1139.02 \text{ (nc)} \checkmark$$